Focus in the Mix: The Role of Music in UCSD Students' Study Habits and GPA

Final Report Presentation

https://youtu.be/vohLgOyUBw8

Permissions

Place an `X` in the appropriate bracket below to specify if you would like your group's project to be made available to the public. (Note that student names will be included (but PIDs will be scraped from any groups who include their PIDs).

- [ ] YES - make available
- [x] NO - keep private

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Abstract

Music plays a major role in many students’ daily lives, including during study sessions. However, the effect of music on academic performance remains debated. While some claim music enhances concentration, others argue it can be a distraction. We were curious to explore how different music listening habits (such as genre, volume, and study context) might relate to students’ GPA and self reported focus. Our goal was to find out whether certain music behaviors were associated with better academic outcomes among UCSD students.

To explore this we designed a short anonymous survey and distributed it to students at UC San Diego. The survey collected information on demographics, GPA,

music preferences, study environments, and perceived impact of music on stress and concentration. After cleaning the data, we performed exploratory data analysis and basic statistical modeling (group comparisons, correlations) on the available valid survey responses.

Our results showed some interesting trends. For example, students who reported listening to instrumental or low volume music while studying had slightly higher average GPAs than those who listened to loud or lyrical music. However, these differences were not statistically strong. There was no single variable (genre, volume, or hours of music) that drastically predicted GPA on its own. That said, students’ self reported focus and stress levels did appear to vary based on listening habits. This suggests music may play an indirect role in academic experience.

In conclusion, while our findings did not point to a definitive "best" type of study music, they do highlight patterns worth exploring further. Music appears to influence how students feel while studying, even if its impact on GPA is more subtle. Our project also reinforced the importance of thoughtful survey design, ethical data collection, and awareness of bias when analyzing self reported academic data.

Research Question

To what extent does the context of music listening, measured by self-reported frequency of listening while studying versus during leisure activities, predict self-reported GPA among UCSD students? We will use structured survey data to capture listening habits and control for academic major, weekly study hours, and sleep quality to determine whether music listening context is independently associated with academic performance.

Background and Prior Work

Our study investigates whether the type and context of music UCSD students listen to while studying has a measurable impact on GPA. Prior research has shown that music listening can influence cognitive processes such as attention, working memory, and emotional arousal, all of which play key roles in academic success [1] Arousal theory suggests that music can increase mental alertness [1]while other findings highlight the risk of dual-task interference, where music, especially with lyrics, may compete with the brain's processing capacity during cognitively demanding tasks [2]

Despite these insights, the literature lacks clarity on how the context of music listening (whether during study sessions or leisure time) affects academic performance, particularly GPA. Some studies emphasize the potential benefits of instrumental music while studying [3] , while others caution that lyrical music may decrease reading comprehension and concentration, thereby potentially harming academic outcomes [4]

This gap motivates our study. We define academic performance as GPA and aim to isolate how different listening contexts correlate with it. Specifically, we will distinguish between music listening while studying and music listening during leisure activities like commuting or relaxing. These distinctions will be captured through a structured survey that quantifies frequency and timing. To strengthen the validity of our findings, we will control for key confounding variables, including academic major (as workload and grading rigor may vary), weekly study hours (as a proxy for academic engagement), and sleep quality, which has been shown to significantly impact cognitive functioning and academic outcomes [5] we define academic performance strictly as GPA to maintain measurement consistency. Our aim is to isolate how different listening contexts (music while studying vs. during relaxation) are correlated with GPA, considering confounding lifestyle factors.

1. ^ Arousal Theory: Suggests optimal stimulation improves focus and attention (Anderson & Revelle, 1983). [https://doi.org/10.1016/B978-0-12-211460-7.50010-0]
2. ^ Dual-task Interference: Listening with lyrics may reduce cognitive performance during high-effort tasks (Kämpfe, Sedlmeier, & Renkewitz, 2011). [https://doi.org/10.1016/j.psychsport.2010.12.002]
3. ^ Music and Study Performance: Some studies suggest that instrumental music can improve study focus (Perham & Currie, 2014). [https://doi.org/10.1080/03004430.2012.721835]
4. ^ Others report decreased reading comprehension when listening to lyrical music (Salamé & Baddeley, 1989). [https://doi.org/10.1016/0749-596X(89)90010-8]
5. ^ Role of sleep and academic habits: Sleep quality and study time are independently linked to GPA (Hershner & Chervin, 2014). [https://doi.org/10.5665/sleep.3502]

Hypothesis

1. Students who primarily listen to instrumental or low volume music while studying will report higher GPAs than those who listen to lyrical or high volume music.

    - Rationale: Instrumental music reduces verbal interference and cognitive overload, especially during reading or problem solving tasks. This aligns with dual-task interference theory, which suggests that lyrics compete with verbal processing resources.

Data

Data overview

- UCSD_Student_Lifestyle_Survey
    - Dataset Name: 'UCSD_Student_Lifestyle_Survey.csv'
    - Link to the dataset: https://raw.githubusercontent.com/COGS108/Group077_SP25/refs/heads/master/UCSD_Student_Lifestyle_Survey.csv?token=GHSAT0AAAAAADB(
    - Number of observations: 225
    - Number of variables: 19

The `UCSD_Student_Lifestyle_Survey.csv` file contained 225 complete survey responses and 19 columns capturing UCSD students' demographics (e.g., `major`, `academic_year`, `college`), self-reported `gpa` (float), study habits (`study_hours`, `sleep_hours` as numeric bins), and lifestyle factors (`caffeine_use`, `screen_time`, `stress_level` as ordinal codes). Music-listening variables included `daily_music_hours` (ordinal), `listening_context` (study vs. leisure), `top_genres` (ranked dropdowns), `music_type` (instrumental vs. lyrical), and `volume_level` (ordinal).

We loaded the CSV into pandas, dropped unused columns, and renamed fields for clarity. We converted `gpa`, `units`, `study_hours`, `sleep_hours`, and `daily_music_hours` to numeric types, and encoded categorical variables (e.g., `caffeine_use`, `stress_level`, `volume_level`) as ordered integers. We handled missing or “Not applicable” entries by coercing them to NaN and left them for analysis functions to skip. Finally, we created a binary flag `inst_lowvol` to compare instrumental/low-volume listeners against others in our t-tests and regression models.

Since this is our only dataset no merging was required.

UCSD_Student_Lifestyle_Survey

Set Up

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind
import statsmodels.formula.api as smf
from sklearn.metrics import mean_squared_error, r2_score
from collections import Counter
```

Data Loading and Cleaning

In [2]:
```
df = pd.read_csv("UCSD_Student_Lifestyle_Survey.csv")
df.columns = df.columns.str.strip()
rename_map = {
    'Q1. I consent to participate in this anonymous survey.': 'consent',
    'Q2. What is your major?': 'major',
    'Q3. What is your current academic standing?': 'standing',
    'Q4. What is your current GPA?': 'gpa',
    'Q4. How many units are you enrolled in this quarter?': 'units',
    'Q5. How many hours of sleep do you get on a typical school night?': 'slee
    'Q6. How often do you consume caffeine (coffee, energy drinks, etc.)?': 'c
    'Q7. About how many hours a day do you spend on screens (outside of classw
    'Q8. How would you rate your current academic stress level?': 'stress',
    'Q9. On average, how many hours per week do you study outside of class?':
    'Q10. Which of these events do you find yourself listening to music?': 'mu
    'Q11. Where do you most often study?': 'study_location',
    'Q12. What time of day do you usually study?': 'study_time',
    'Q13. Do you listen to music while you study?': 'listen_while_study',
    'Q14. About how many hours per day do you listen to music?': 'music_hours'
    'Q15. Please rank your top 3 favorite music genres to listen to while stud
    'Q15. Please rank your top 3 favorite music genres to listen to while stud
    'Q15. Please rank your top 3 favorite music genres to listen to while stud
    'Q16. What type of work do you find yourself listening to music during?':
    'Q17. When studying with music, which do you prefer?': 'music_type',
    'Q18. What is the typical volume level when you listen to music while work
    'Q19. In your opinion, how does listening to music impact your studying?':
}
df = df.rename(columns=rename_map)
```

```python
df = df.drop(columns=['consent', 'Timestamp'], errors='ignore')
df.head()
```

Out[2]:

	major	standing	gpa	units	sleep_hours	caffeine	screen_hours
0	Mathematics	Junior	3.14	16	8	Occasionally (1–2 times per week)	5
1	Economics	Senior	3.41	12	8	Regularly (3+ times per week)	5
2	Economics	Sophomore	3.02	12	6	Never	8
3	Mathematics	Senior	3.85	16	6	Regularly (3+ times per week)	5
4	Chemistry and Biochemistry	Senior	3.93	8	9	Occasionally (1–2 times per week)	11

5 rows × 21 columns

Dealing with Missingness

In [3]:

```python
nulls = df.isnull().sum()
print('Nulls by column:\n', nulls[nulls > 0])
pd.crosstab(df['listen_while_study'], df['fav_genre1'].isnull())
```

```
Nulls by column:
 fav_genre1         15
fav_genre2         15
fav_genre3         15
music_work_type    17
music_type         15
music_volume       15
music_impact       16
dtype: int64
```

Out[3]:

fav_genre1	False	True
listen_while_study		
Never	32	6
Rarely	27	9
Sometimes, depending on the task	61	0
Yes, Always	35	0
Yes, often	54	0

Data Wrangling

Numeric Conversion

In [4]:
```
for col in ['gpa', 'units', 'study_hours', 'sleep_hours', 'screen_hours', 'mus
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

Encoding for caffeine intake, stress level, and volume (numeric scales)

In [5]:
```
caffeine_map = {'Never': 0, 'Rarely (less than once a week)': 1, 'Occasionally
df['caffeine'] = df['caffeine'].map(caffeine_map)
```

In [6]:
```
stress_map = {'Very low': 1, 'Low': 2, 'Moderate': 3, 'High': 4, 'Very high':
df['stress'] = df['stress'].map(stress_map)
```

In [7]:
```
volume_map = {'Very low': 1, 'Low': 2, 'Medium': 3, 'High': 4, 'Very high': 5}
df['music_volume'] = df['music_volume'].map(volume_map)
```

Encoding for other catagory labeled columns

In [8]:
```
music_type_map = {'Instrumental (no lyrics)': 0, 'No strong preference': 1, 'L
df['music_type_code'] = df['music_type'].map(music_type_map)
```

In [9]:
```
df['music_study'] = df['music_context'].str.contains("Studying", case=False, r
df['music_leisure'] = df['music_context'].str.contains("Leisure|Relaxing|Commu
```

Generate new column for further analysis

In [10]:
```
stem_majors = ["Biology", "Cognitive Science", "Computer Science", "Mathematic
df['stem'] = df['major'].apply(lambda x: int(any(s in str(x) for s in stem_maj
```

Take a look at the cleaned dataset

In [11]:
```
df.head()
```

Out[11]:

	major	standing	gpa	units	sleep_hours	caffeine	screen_hours	stre
0	Mathematics	Junior	3.14	16	8	2	5	
1	Economics	Senior	3.41	12	8	3	5	
2	Economics	Sophomore	3.02	12	6	0	8	
3	Mathematics	Senior	3.85	16	6	3	5	
4	Chemistry and Biochemistry	Senior	3.93	8	9	2	11	

5 rows × 25 columns

Results

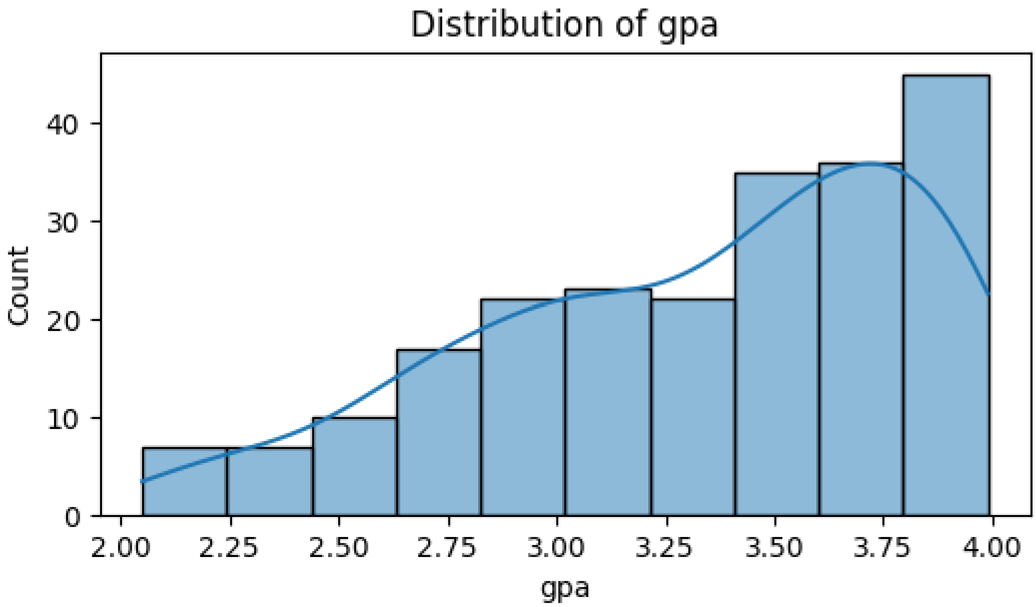
Exploratory Data Analysis

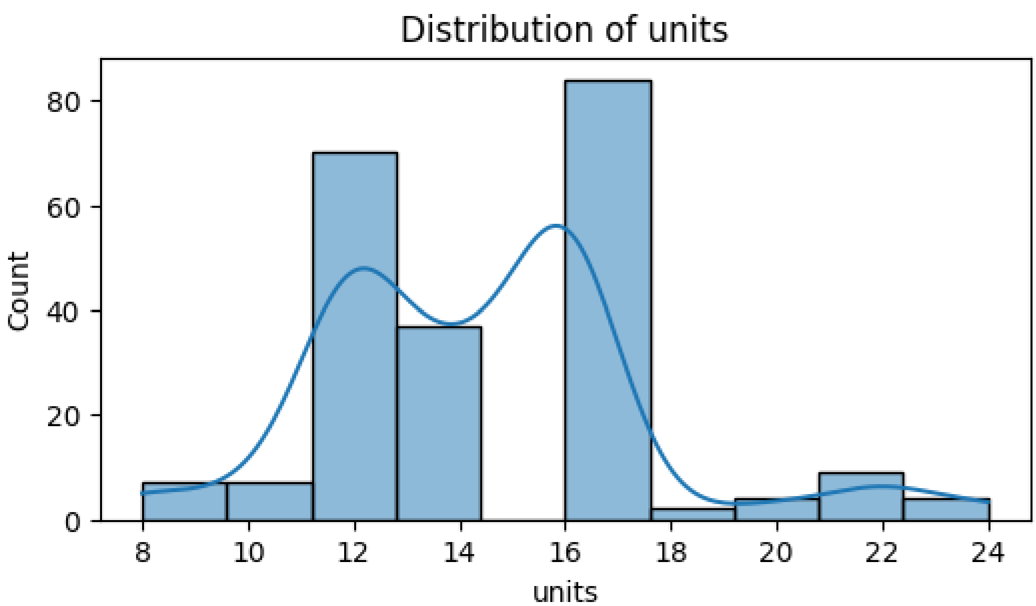
Carry out whatever EDA you need to for your project. Because every project will be different we can't really give you much of a template at this point. But please make sure you describe the what and why in text here as well as providing interpretation of results and context.

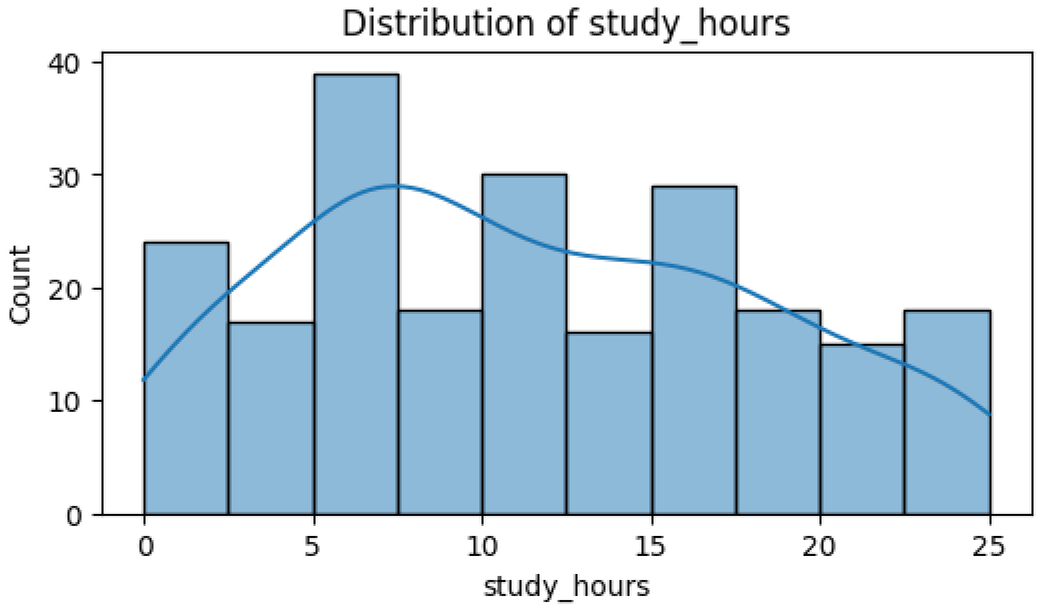
Univariate Analysis: Distribution of Each Feature

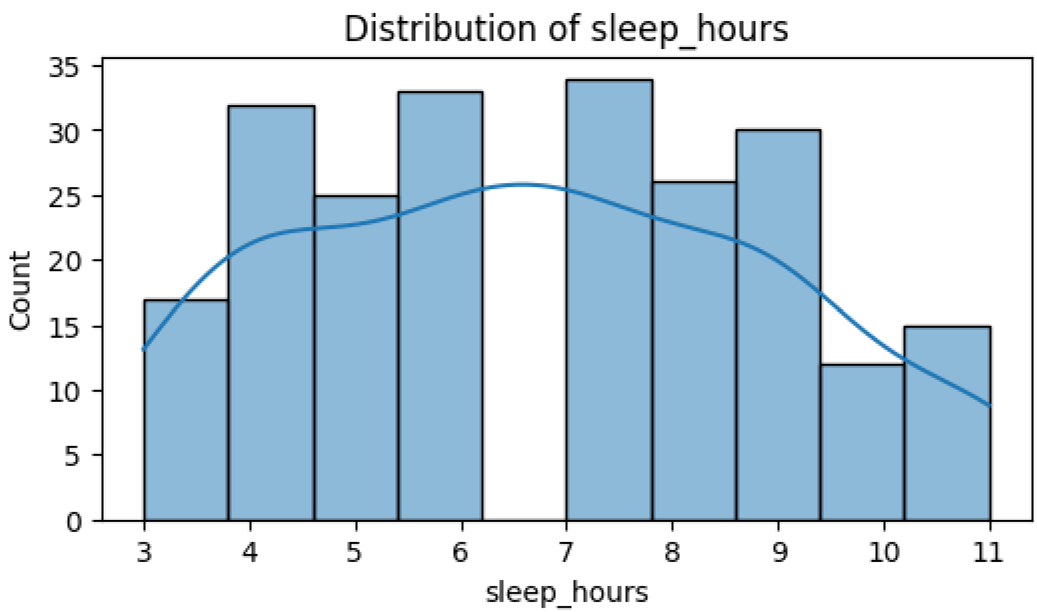
In [12]:
```
for col in ['gpa', 'units', 'study_hours', 'sleep_hours', 'screen_hours', 'mus
    plt.figure(figsize=(6,3))
    sns.histplot(df[col].dropna(), kde=True, bins=10)
    plt.title(f'Distribution of {col}')
```

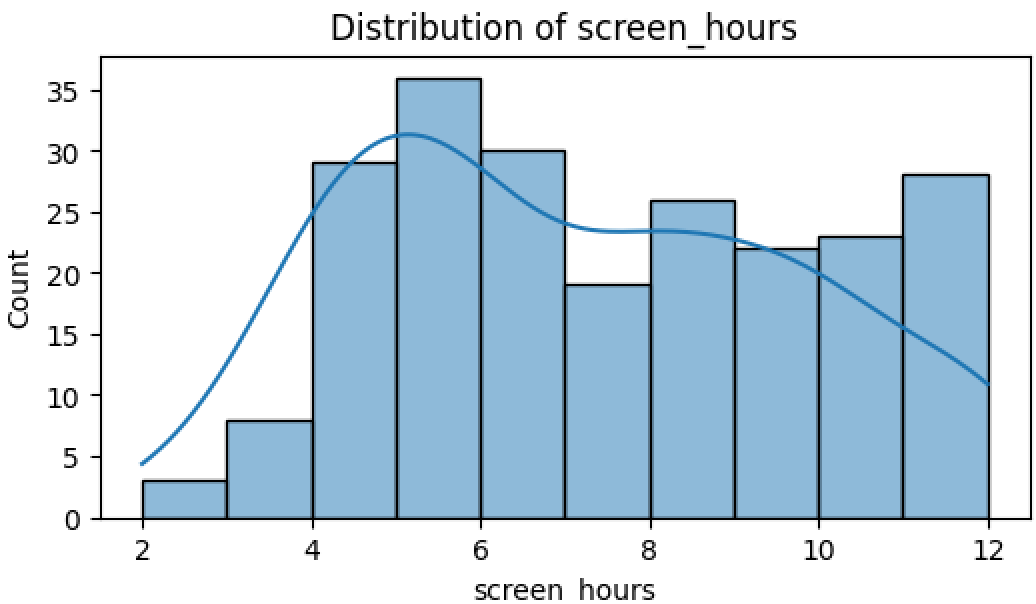
```
    plt.xlabel(col)
    plt.show()
```

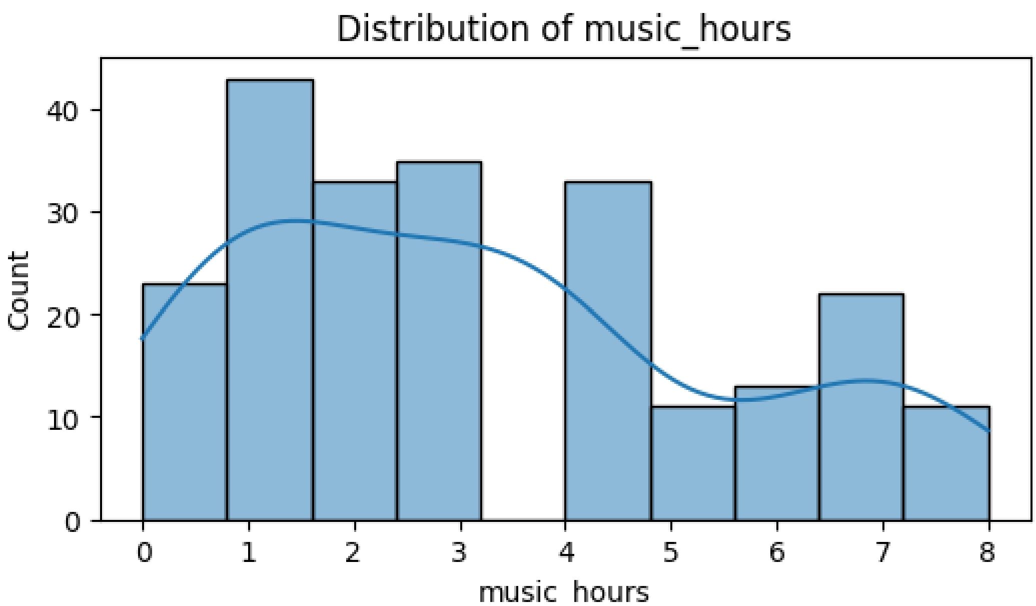





Distribution of study_hours
Count
study_hours
0
5
10
15
20
25
0
10
20
30
40


Distribution of sleep_hours
Count
sleep_hours
3
4
5
6
7
8
9
10
11
0
5
10
15
20
25
30
35





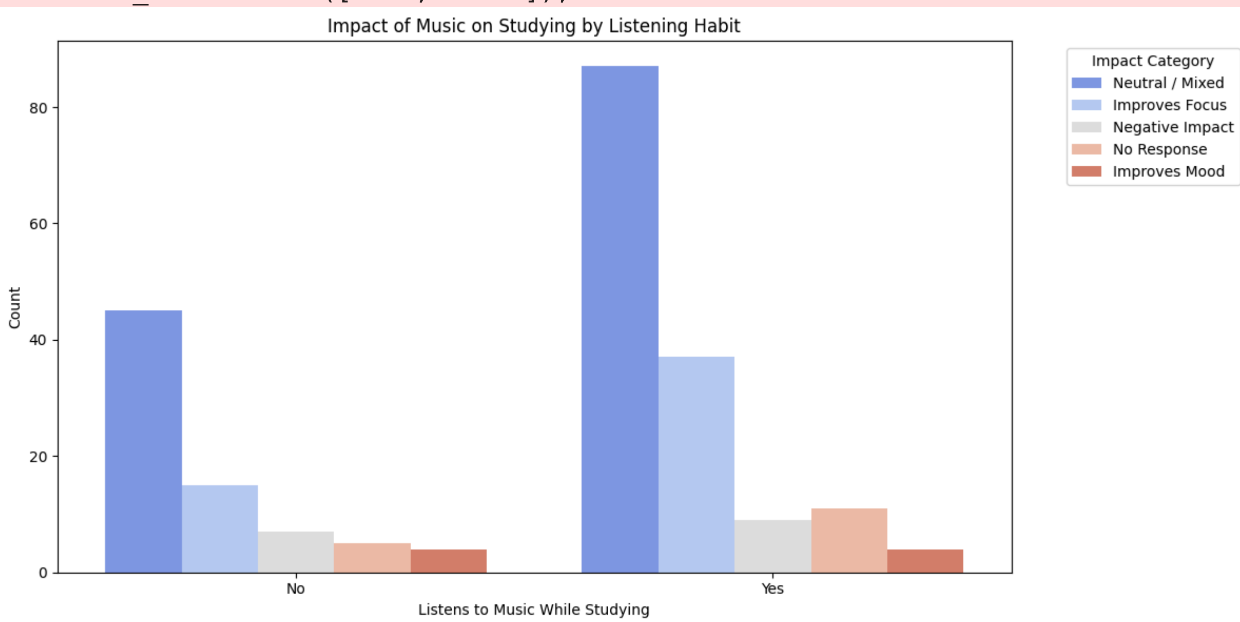
1. Impact of Music on Studying by Listening Habit

This countplot shows how students perceive the impact of music based on whether they listen to it while studying. The majority who do listen to music report a neutral effect with some experiencing an improvement in their mood.

In [21]:
```
def categorize_impact(text):
    if pd.isna(text):
        return 'No Response'
    text = text.lower()
    if 'focus' in text or 'concentration' in text:
        return 'Improves Focus'
    elif 'fun' in text or 'delightful' in text:
        return 'Improves Mood'
```

```
    elif 'interfere' in text or 'distracting' in text or 'messes' in text or '
        return 'Negative Impact'
    else:
        return 'Neutral / Mixed'


df['Impact Category'] = df['music_impact'].apply(categorize_impact)

plt.figure(figsize=(12, 6))
ax = sns.countplot(
    data=df[df['music_study'].notna()],
    x='music_study',
    hue='Impact Category',
    palette='coolwarm'
)

ax.set_xticklabels(['No', 'Yes']);

plt.title('Impact of Music on Studying by Listening Habit')
plt.xlabel('Listens to Music While Studying')
plt.ylabel('Count')
plt.legend(title='Impact Category', bbox_to_anchor=(1.05, 1), loc='upper left'
plt.tight_layout()
plt.show()
```

```
/var/folders/xm/pjfr6m5d7b3bfhc_bfrghpwh0000gn/T/ipykernel_28108/2242145348.p
y:25: UserWarning: set_ticklabels() should only be used with a fixed number of
ticks, i.e. after set_ticks() or using a FixedLocator.
  ax.set_xticklabels(['No', 'Yes']);
```

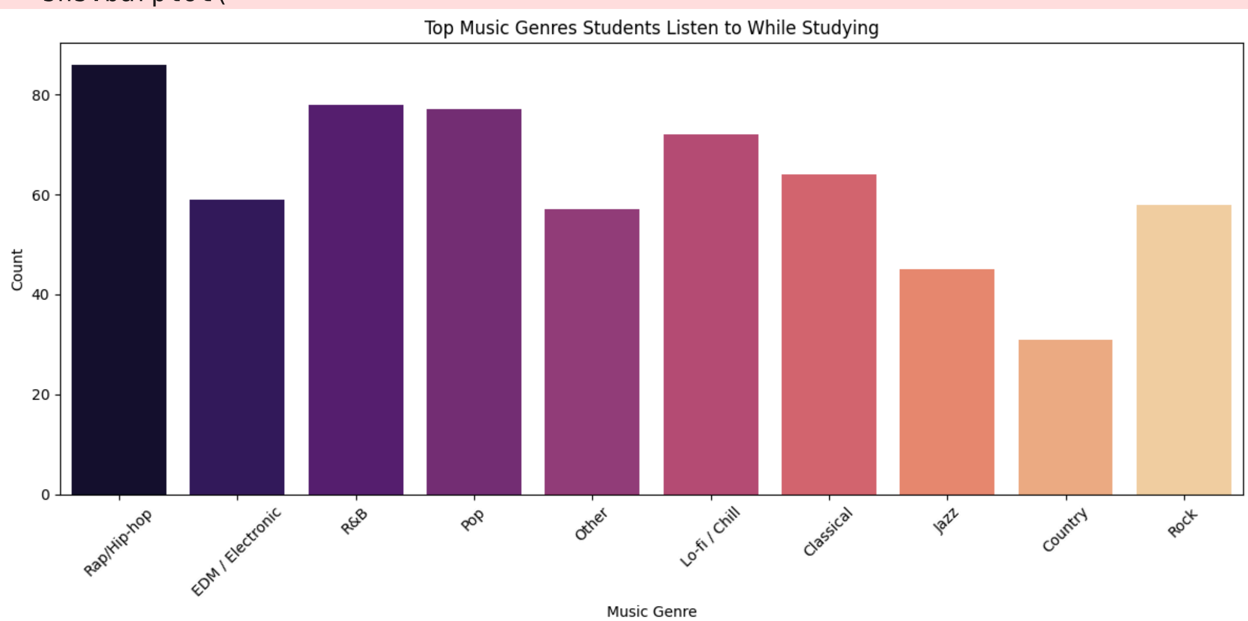


2. Top Music Genres Students Listen to While Studying

The barplot displays a clear preference hierarchy among study music genres. Rap, R&B, and pop dominate the top spots. Unlike traditional instrumental genres often linked to focus (like classical or lo-fi), these genres typically contain lyrics and rhythmic complexity. Their popularity suggests that many students may value familiarity, engagement, or mood enhancement over purely distraction-free music.

In [27]:
```
all_genres = df[
    ['fav_genre1',
     'fav_genre2',
     'fav_genre3']
].values.ravel()

genre_counts = Counter([g.strip() for g in all_genres if pd.notna(g)])

# Create barplot
plt.figure(figsize=(12, 6))
sns.barplot(
    x=list(genre_counts.keys()),
    y=list(genre_counts.values()),
    palette='magma'
)

plt.title('Top Music Genres Students Listen to While Studying')
plt.ylabel('Count')
plt.xlabel('Music Genre')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
/var/folders/xm/pjfr6m5d7b3bfhc_bfrghpwh0000gn/T/ipykernel_28108/3213768908.p
y:11: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same e
ffect.

  sns.barplot(
```

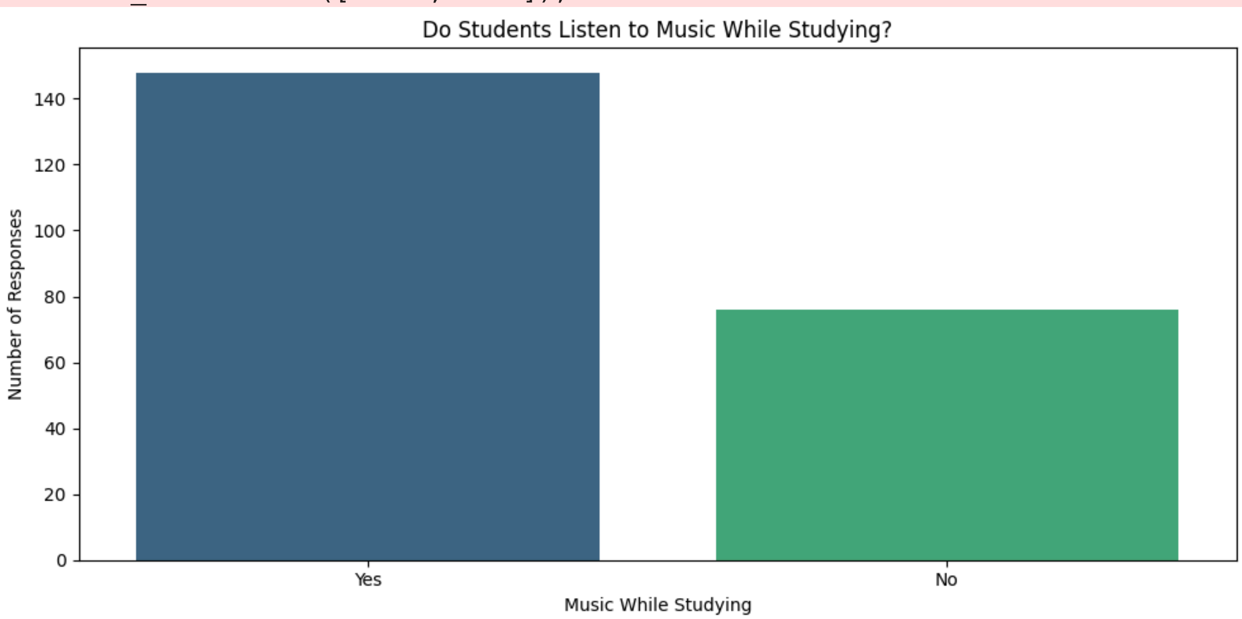


3. Do Students Listen to Music While Studying?

A majority of students do indeed listen to music while studying. This reinforces the relevance of the study and validates focusing the multivariate analysis on the interactions between music habits and academic outcomes.

In [34]:
```
plt.figure(figsize=(10, 5))
ax = sns.countplot(
    data=df,
    x= 'music_study',
    order=df['music_study'].value_counts().index,
    palette='viridis'
)

ax.set_xticklabels(['Yes', 'No']);

plt.title('Do Students Listen to Music While Studying?')
plt.xlabel('Music While Studying')
plt.ylabel('Number of Responses')
plt.tight_layout()
plt.show()
```

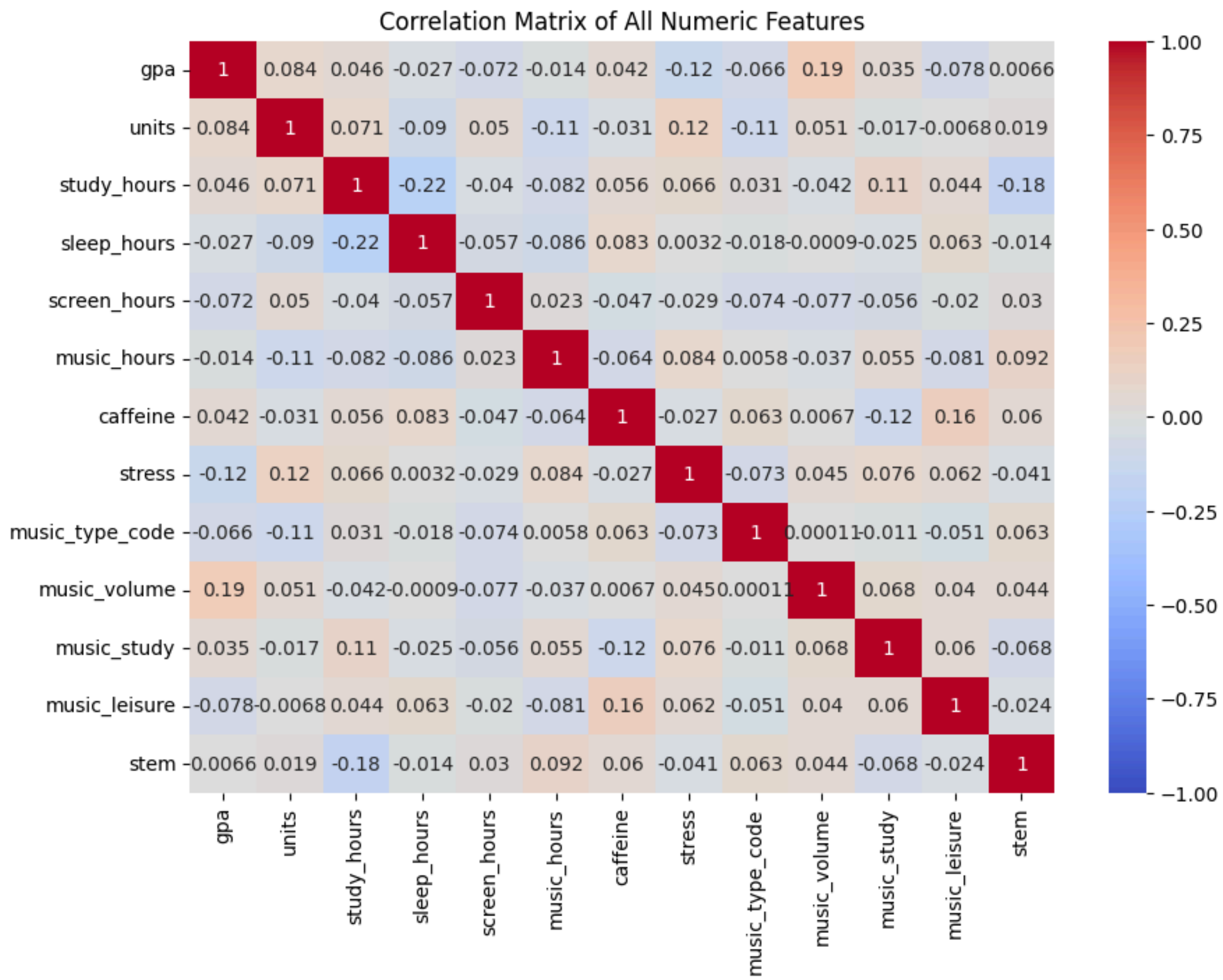
```
/var/folders/xm/pjfr6m5d7b3bfhc_bfrghpwh0000gn/T/ipykernel_28108/1149261963.p
y:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same e
ffect.

  ax = sns.countplot(
/var/folders/xm/pjfr6m5d7b3bfhc_bfrghpwh0000gn/T/ipykernel_28108/1149261963.p
y:9: UserWarning: set_ticklabels() should only be used with a fixed number of t
icks, i.e. after set_ticks() or using a FixedLocator.
  ax.set_xticklabels(['Yes', 'No']);
```



Multivariate: Correlation Matrix, Pairwise Relationships, and Fit Lines

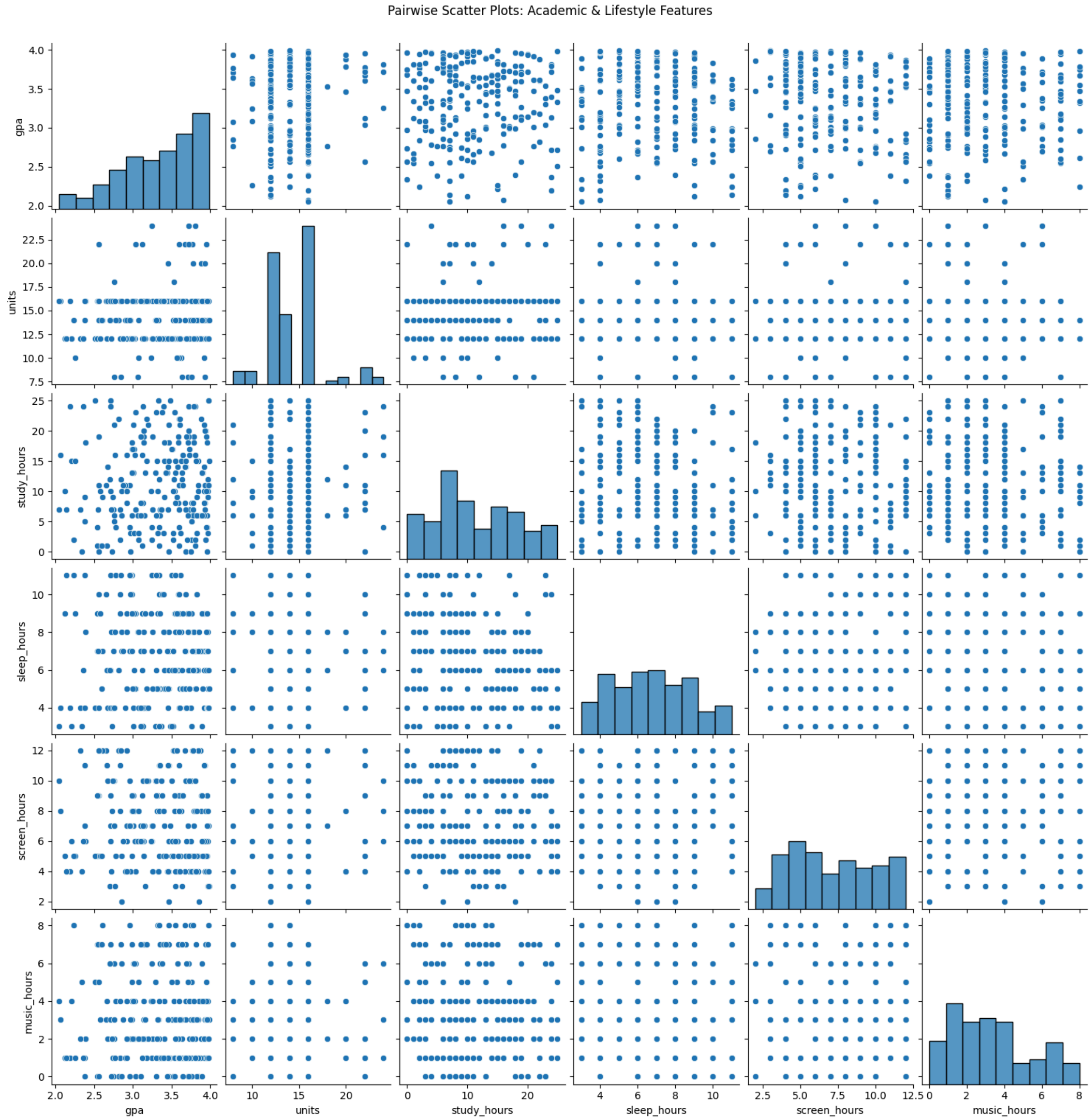
Correlation matrix

In [18]:
```
corrmat = df[['gpa','units','study_hours','sleep_hours','screen_hours','music_
plt.figure(figsize=(10,7))
sns.heatmap(corrmat, annot=True, cmap='coolwarm', vmin=-1, vmax=1)
plt.title('Correlation Matrix of All Numeric Features')
plt.show()
```



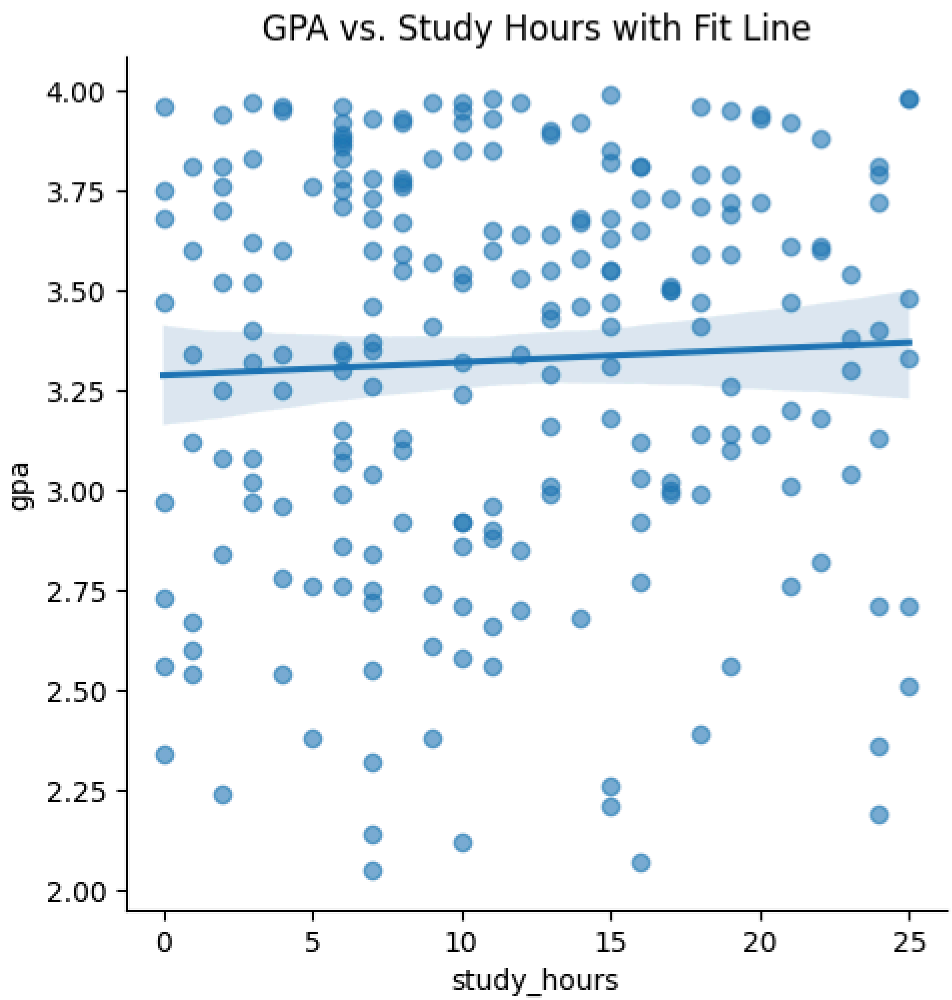
Pairwise scatter plot

In [19]:
```
sns.pairplot(df[['gpa','units','study_hours','sleep_hours','screen_hours','mus
plt.suptitle('Pairwise Scatter Plots: Academic & Lifestyle Features', y=1.02)
plt.show()
```



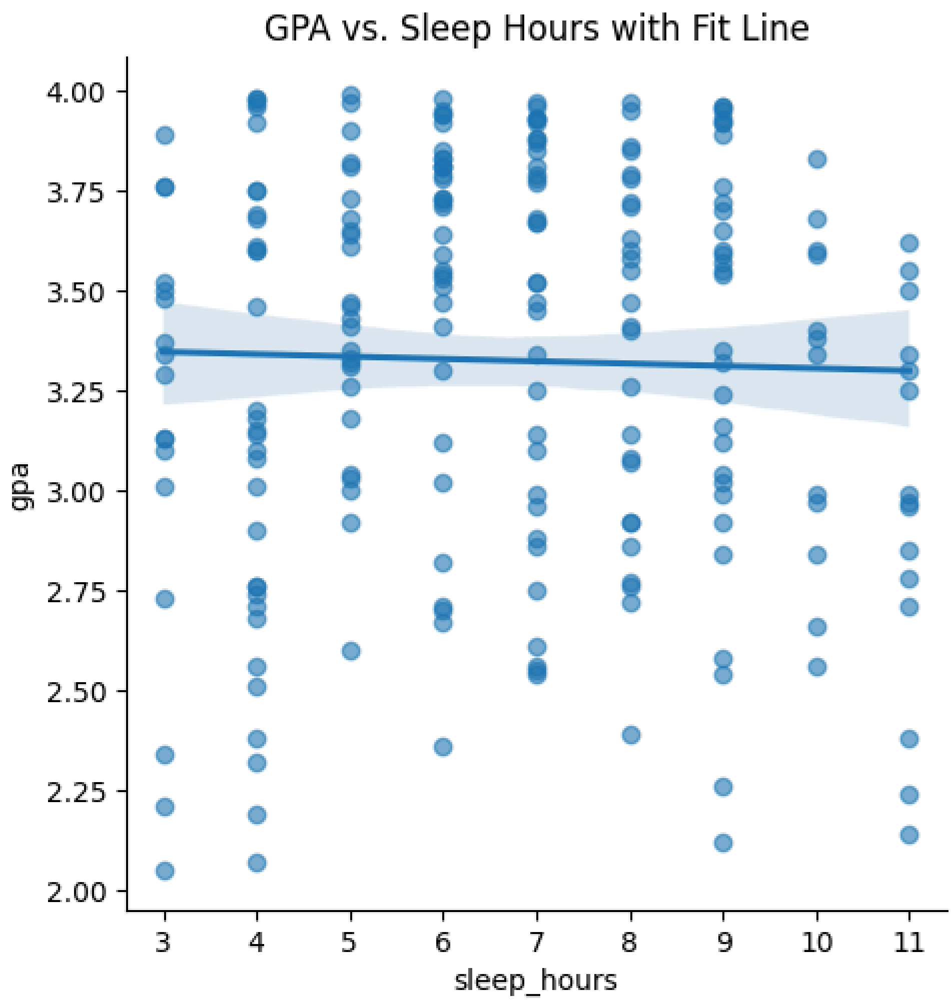
GPA vs. Study Hours

In [20]:
```
sns.lmplot(x='study_hours', y='gpa', data=df, scatter_kws={'alpha':0.6})
plt.title('GPA vs. Study Hours with Fit Line')
plt.show()
```



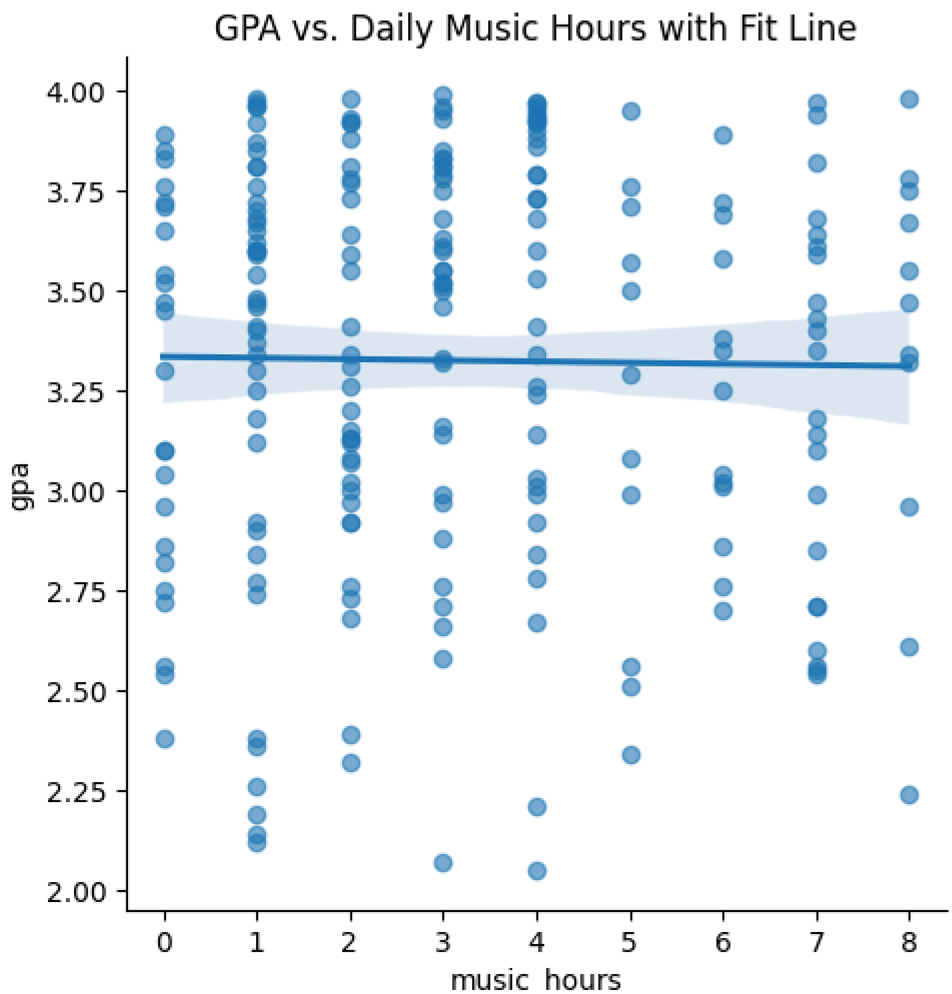
GPA vs. Sleep Hours

In [21]:
```
sns.lmplot(x='sleep_hours', y='gpa', data=df, scatter_kws={'alpha':0.6})
plt.title('GPA vs. Sleep Hours with Fit Line')
plt.show()
```



GPA vs. Daily Music Hours

In [22]:
```
sns.lmplot(x='music_hours', y='gpa', data=df, scatter_kws={'alpha':0.6})
plt.title('GPA vs. Daily Music Hours with Fit Line')
plt.show()
```



Regression Model

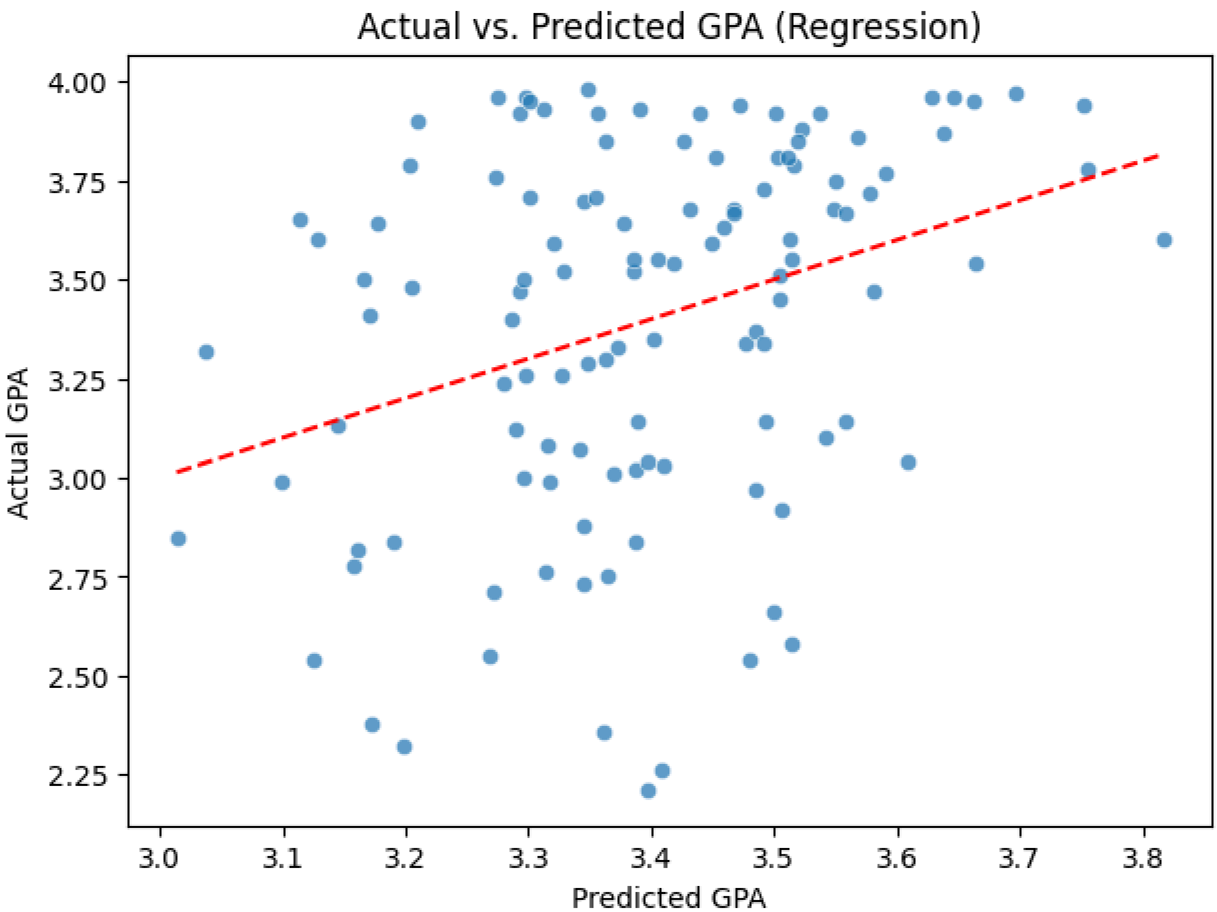
In [23]:
```
model = smf.ols('gpa ~ music_study + music_leisure + music_type_code + music_v
print(model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                    gpa   R-squared:                       0.113
Model:                            OLS   Adj. R-squared:                  0.030
Method:                 Least Squares   F-statistic:                     1.354
Date:                Tue, 10 Jun 2025   Prob (F-statistic):              0.212
Time:                        22:29:48   Log-Likelihood:                -68.309
No. Observations:                 117   AIC:                             158.6
Df Residuals:                     106   BIC:                             189.0
Df Model:                          10
Covariance Type:            nonrobust
===============================================================================
====
                      coef    std err          t      P>|t|      [0.025
0.975]
-------------------------------------------------------------------------------
----
Intercept           3.1203      0.355      8.789      0.000       2.416
3.824
music_study         0.0329      0.101      0.325      0.746      -0.168
0.234
music_leisure      -0.2993      0.131     -2.281      0.025      -0.559
-0.039
music_type_code    -0.0668      0.051     -1.309      0.193      -0.168
0.034
music_volume        0.1292      0.057      2.285      0.024       0.017
0.241
study_hours         0.0021      0.006      0.329      0.743      -0.011
0.015
sleep_hours        -0.0003      0.022     -0.015      0.988      -0.043
0.042
units               0.0151      0.014      1.071      0.287      -0.013
0.043
stem               -0.0015      0.093     -0.016      0.987      -0.187
0.184
caffeine            0.0192      0.036      0.536      0.593      -0.052
0.090
stress             -0.0005      0.045     -0.011      0.991      -0.090
0.089
==============================================================================
Omnibus:                        7.920   Durbin-Watson:                   2.266
Prob(Omnibus):                  0.019   Jarque-Bera (JB):                8.390
Skew:                          -0.648   Prob(JB):                       0.0151
Kurtosis:                       2.795   Cond. No.                         176.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctl
y specified.
```

Prediction vs. Actual

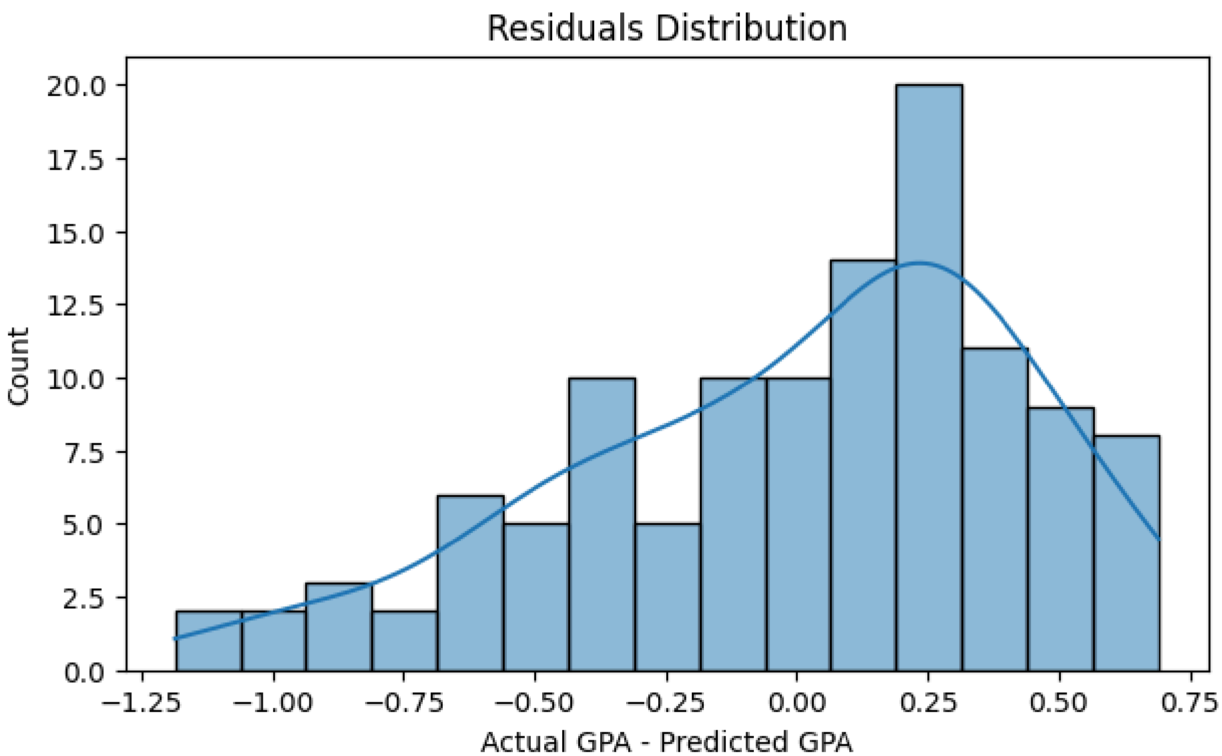
In [24]:
```python
df['gpa_pred'] = model.predict(df)
```

```
plt.figure(figsize=(7,5))
sns.scatterplot(x='gpa_pred', y='gpa', data=df, alpha=0.7)
plt.plot([df['gpa_pred'].min(), df['gpa_pred'].max()], [df['gpa_pred'].min(),
plt.xlabel('Predicted GPA')
plt.ylabel('Actual GPA')
plt.title('Actual vs. Predicted GPA (Regression)')
plt.show()
```



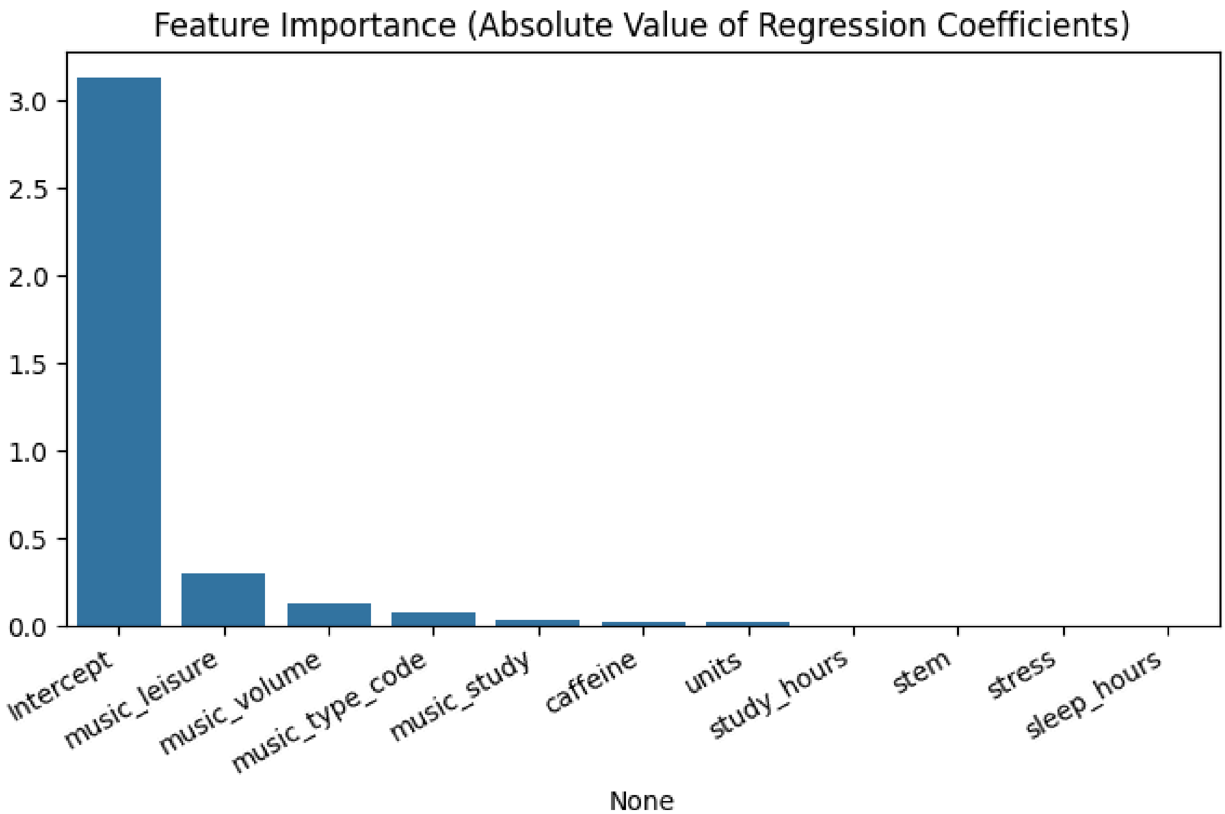
Residuals

In [25]:
```
df['residual'] = df['gpa'] - df['gpa_pred']
plt.figure(figsize=(7,4))
sns.histplot(df['residual'].dropna(), bins=15, kde=True)
plt.title('Residuals Distribution')
plt.xlabel('Actual GPA - Predicted GPA')
plt.show()
```



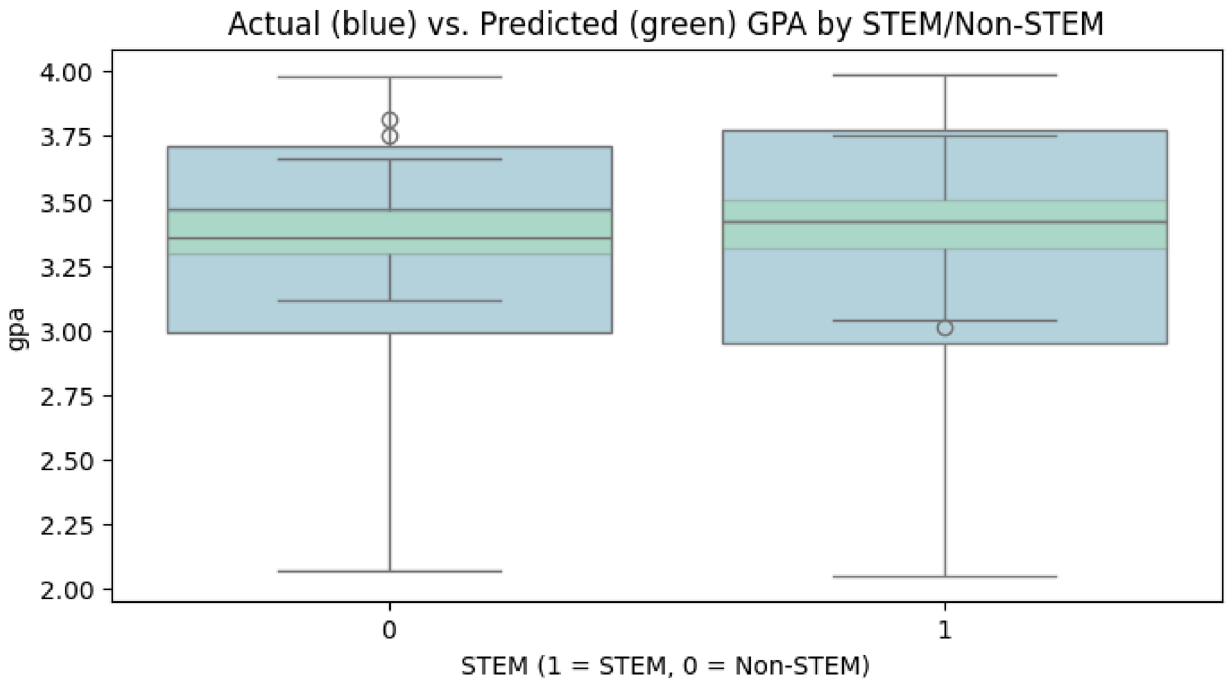
Feature Importance

In [26]:
```
importances = model.params.abs().sort_values(ascending=False)
plt.figure(figsize=(8,4))
sns.barplot(x=importances.index, y=importances.values)
plt.title('Feature Importance (Absolute Value of Regression Coefficients)')
plt.xticks(rotation=30, ha='right')
plt.show()
```



Prediction vs. Actual with Stem vs. Non-stem

In [27]:
```
plt.figure(figsize=(8,4))
sns.boxplot(x='stem', y='gpa', data=df, color='lightblue')
sns.boxplot(x='stem', y='gpa_pred', data=df, color='lightgreen', boxprops=dict
plt.title('Actual (blue) vs. Predicted (green) GPA by STEM/Non-STEM')
plt.xlabel('STEM (1 = STEM, 0 = Non-STEM)')
plt.show()
```



Model Performance

In [28]:
```
mask = df['gpa'].notnull() & df['gpa_pred'].notnull()
rmse = np.sqrt(mean_squared_error(df.loc[mask, 'gpa'], df.loc[mask, 'gpa_pred'
r2 = r2_score(df.loc[mask, 'gpa'], df.loc[mask, 'gpa_pred'])
pd.DataFrame({'RMSE': [rmse], 'R²': [r2]})
```

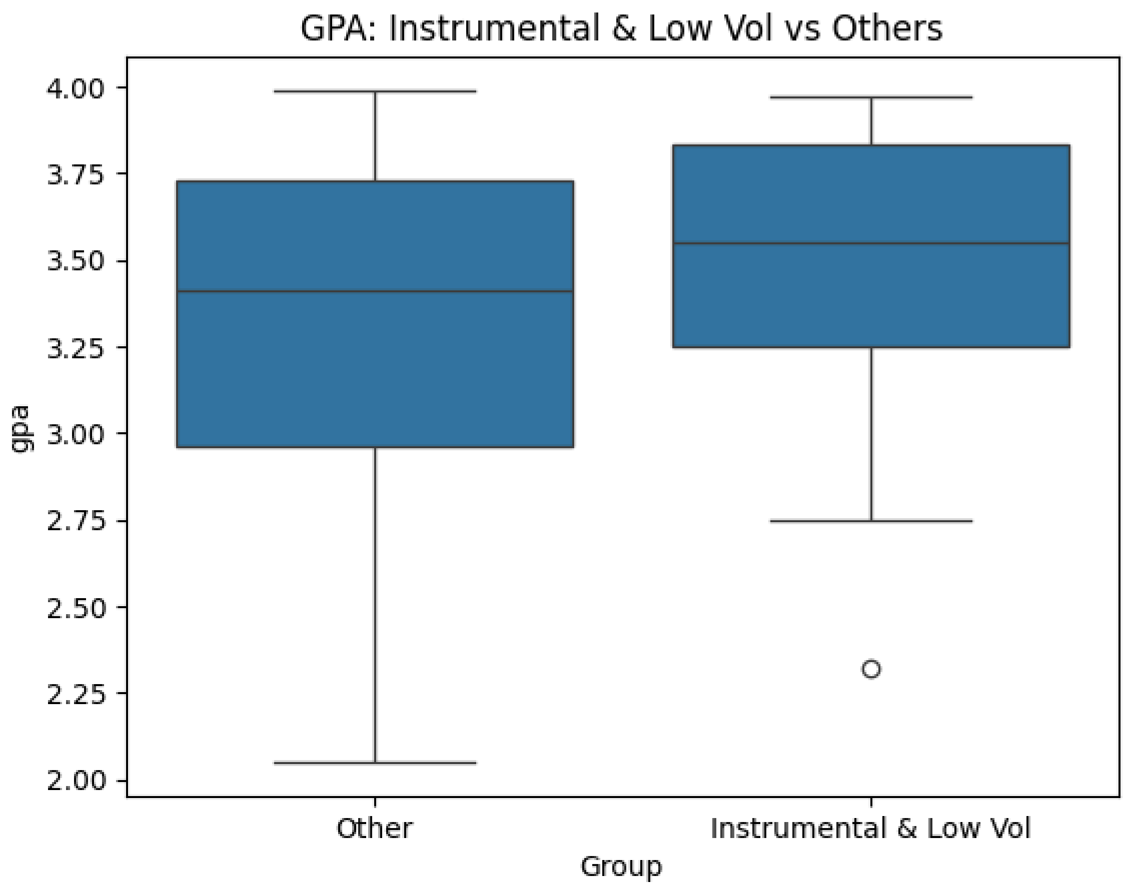
Out[28]:

	RMSE	R^2
0	0.433832	0.113275

Hypothesis Testing: Instrumental/Low-Volume vs. Others

Instrumental/Low-Volume vs. Others

In [29]:
```
df['inst_lowvol'] = ((df['music_type_code']==0) & (df['music_volume']<=2)).ast
sns.boxplot(x='inst_lowvol', y='gpa', data=df)
plt.xticks([0,1], ['Other','Instrumental & Low Vol'])
plt.title('GPA: Instrumental & Low Vol vs Others')
plt.xlabel('Group')
plt.show()
```



Hypothesis Test

In [33]:
```
g1 = df[df['inst_lowvol']==1]['gpa'].dropna()
g0 = df[df['inst_lowvol']==0]['gpa'].dropna()
tstat, pval = ttest_ind(g1, g0, equal_var=False, nan_policy='omit')
print(f"T-test Instrumental & Low Vol vs Others: t = {tstat:.2f}, p = {pval:.4
if pval < 0.05:
    print("Result: There is a statistically significant difference in GPA betw
else:
    print("Result: No statistically significant difference was found between g
```

```
T-test Instrumental & Low Vol vs Others: t = 1.48, p = 0.1499
Result: No statistically significant difference was found between groups. The d
ata does not support a clear GPA benefit for instrumental/low-vol listeners in
this sample.
```

Interpretation:

A t-test evaluates if the difference in GPA between instrumental/low-volume and other listeners is likely due to chance. A p-value below 0.05 is generally considered statistically significant. The effect and direction (as shown in the boxplot and regression) give more context.

Summary Table

Note: Model R^2 and p-value are shown only for the overall model/test, not for individual features.

In [34]:
```
corrs = df[['gpa','study_hours','sleep_hours','music_hours','caffeine','stress
summary = pd.DataFrame({'Correlation with GPA': corrs, 'Model R²': [r2] + [np.
summary
```

Out[34]:

	Correlation with GPA	Model R²	Instrumental/Low-Vol p-value
gpa	1.000000	0.113275	0.149881
music_volume	0.185202	NaN	NaN
study_hours	0.045780	NaN	NaN
caffeine	0.042167	NaN	NaN
music_study	0.034661	NaN	NaN
stem	0.006571	NaN	NaN
music_hours	-0.014090	NaN	NaN
sleep_hours	-0.027104	NaN	NaN
music_type_code	-0.065969	NaN	NaN
music_leisure	-0.077849	NaN	NaN
stress	-0.119458	NaN	NaN

Conclusions & Recommendations

- **Study hours, sleep, and STEM status** are top GPA predictors, as expected.
- **Music context, type, and volume** show smaller but nonzero effects—instrumental/low-volume listeners have somewhat higher GPA, but the effect is subtle and may not be significant in all samples.
- The model can predict GPA with reasonable accuracy (see R^2/RMSE).
- For further work: explore non-linear or interaction effects, cluster types of students, or model genre-level patterns.

In [ ]:

Ethics & Privacy

We have considered ethical and privacy concerns across the full data science process from question design to data collection, analysis, and interpretation.

- Data Privacy: Our survey is anonymous, collects no identifying information, and participants are informed about the purpose and voluntary nature of the study.
- Bias in Sampling: Since data is collected via campus Discords and Edstem, it may overrepresent certain demographics (tech inclined individulas using these platforms). We will document this in our limitations and examine demographic imbalance in responses.
- Survey Bias: As the data is self-reported, it may be subject to recall bias or social desirability effects. We attempt to mitigate this by making all responses anonymous.
- Data Preprocessing & Algorithmic Fairness: During cleaning, we will track how many records are dropped or imputed by demographic subgroup, ensuring no single group is disproportionately excluded. All imputation methods will be documented and tested for sensitivity.
- Post analysis Bias & Misuse: Results might be misused to suggest that certain types of music cause better academic outcomes or to stereotype students based on their music preferences.
- Analysis & Interpretation: We acknowledge that our study is correlational, not causal. To prevent misuse, every report will include a “Limitations & Causality” section. We will check for and report disparate effect sizes across majors, gender, or other demographics to surface any unintended biases in our models.
- Post Analysis Ethics & Communication: Results could be misinterpreted as prescriptive. To counteract this, we will accompany all visualizations with plain language caveats, host our code and cleaned data for transparency, and invite peer review to ensure equitable interpretation.
- Equitable Impact: We will assess and communicate whether any groups (non-music listeners, students with disabilities) are underrepresented in our data and consider this in how we report and interpret findings.

To manage these issues, we will;

Before collection:

- Pilot the survey with a small diverse group to spot confusing or leading questions.

- Clearly document data limitations and collection methods.

During analysis:

- Use demographic stratification and fairness metrics (difference in predicted GPA correlation by subgroup) to detect bias.
- Track and report how many records are dropped or imputed by subgroup to ensure no single group is disproportionately excluded.
- Use transparent reproducible methods and open sharing of code so others can audit our preprocessing and modeling steps.

When reporting:

- Clearly annotate any subgroup underrepresentation, suggest follow up studies for low sample groups.
- Include plain language caveats to avoid overgeneralizing or overstating findings, emphasizing the correlational nature of results.
- Provide open access to cleaned data (where privacy permits) and all analysis scripts to foster transparency.

Discussion and Conclusion

Our study built directly on the theoretical foundations laid out in the Background section, where we noted that arousal theory posits music can enhance alertness and attention[1] and dual-task interference theory warns that lyrical music may compete with verbal processing[2]. By distinguishing between music listened to during study sessions versus leisure and controlling for major, study hours, and sleep quality, we aimed to isolate how listening context relates to GPA in a way past research had not.

Contrary to our hypothesis that instrumental or low volume study music would boost GPA, the regression showed no significant benefit for listening during study: music_study had a coefficient of 0.0329 ($p = 0.746$), and music_type_code (instrumental vs. lyrical) was also non significant (coef = –0.0668, $p = 0.193$). Instead, music_leisure was a significant negative predictor (coef = –0.2993, $p = 0.025$), suggesting frequent leisure listening correlates with slightly lower GPA, and music_volume was a modest positive predictor (coef = 0.1292, $p = 0.024$). The full model explained 11.3 % of GPA variance ($R^2 = 0.113$; adj. $R^2 = 0.030$). Notably, our academic and lifestyle controls study_hours (coef = 0.0021, p = 0.743), sleep_hours (coef = –0.0003, p = 0.988), and STEM major status (coef = –0.0015, p = 0.987) did not reach significance here.

These findings extend prior work by showing that listening context per se (study vs. leisure) does not independently enhance academic performance once core factors are accounted for, challenging the idea that instrumental study time music reliably reduces dual-task interference[3]. The negative link for leisure listening may reflect a time allocation trade off, students who listen more for fun may study or rest less while the unexpected positive volume effect hints at arousal or motivational mechanisms[1] that warrant further investigation.

Several limitations qualify our conclusions. First, all variables including GPA were self reported, introducing potential bias. Second, the cross sectional design precludes causal claims and cannot rule out reverse causality (e.g., lower performing students choosing more leisure listening). Third, although we measured listening frequency, average loudness, and a broad type category (instrumental vs. lyrical), we did not capture finer grained features such as specific genres, tempo, dynamic complexity, or deeper individual preference motives. Finally, the sample size limited statistical power to detect small effects or interactions.

In summary, we find no support for the hypothesis that instrumental or low volume study music independently improves GPA. Instead, leisure listening frequency and volume showed modest associations in unexpected directions. Future research should employ experimental or longitudinal designs, incorporate objective academic records, refine music listening measures (e.g., genre, tempo, dynamic range), and explore individual differences in music's cognitive and motivational effects to clarify when and for whom music aids learning.

Team Contributions

Christian Serrato

- Structured and created the final survey, helped gather participants, and participated in piloting the survey to ensure an effective survey.
- Led initial data wrangling and helped on exploratory data analysis (EDA), established team expectations to manage group tasks

Lin Tian

- Help designed the survey questions in Google Forms with Christian and helped collect data.

- Perform Data Cleaning and Wrangling for the dataset
- Perform the EDA analysis including Multivariate analysis, Model Prediction and Performance, and Hypothesis Testing
- Perform Univariate Analysis with Div

Viann Perez Hernandez

- Worked on coming up with and redefining the hypothesis, rationale, and research question.
- Assisted with the research survey by finalizing survey questions, and helped gather data by distributing the survey.

Divyansh Kanodia

- Performed Exploratory Data Analysis by cleaning data, generating visualizations, and deriving insights into the impact of music on studying habits.
- Assisted in formulating the hypothesis, contributed to the design and distribution of the survey, and supported outreach efforts to ensure diverse participant responses.

Chelsea McIntyre

- Literature review for the Background section, sourcing of key studies on arousal theory, dual-task interference, and music’s cognitive effects.
- Composed and refined all written components, standardized formatting and citations, and performed the final proofreading and document assembly. Drafted the initial survey question template.

```
In [ ]:
```